An Auditable Multi-Agent System for Hypothesis Generation in C9orf72-Associated ALS

Technical Report

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Abstract

Autonomous scientific discovery requires more than generating plausible hypotheses: scientific claims must remain traceable to evidence, open to criticism, and experimentally falsifiable. Starting from the C9orf72-associated T-cell autoreactivity reported by Michaelis et al. [15], we investigate a central question: **Can a multi-agent system autonomously transform a biomedical observation into auditable, mechanistically explicit, and falsifiable therapeutic hypotheses for amyotrophic lateral sclerosis (ALS)?**

We implement a twelve-phase workflow combining literature retrieval, independent hypothesis generation, adversarial review, evidence verification, computational analysis, experimental design, and therapeutic synthesis. The system builds on *Claude Code*'s dynamic workflows for multi-agent coordination, while a deterministic Python orchestrator controls phase execution, validation, and persistent state. An append-only knowledge base represents scientific claims, provenance, supporting and contradicting relationships, and epistemic status.

We evaluate five end-to-end runs under identical configurations, comparing three executions with an initially empty knowledge base against two using accumulated knowledge. All runs generated structured research proposals; two returned a proposal outcome and three reported insufficient evidence. Deterministic biomedical analyses were reproducible, but agent-generated interpretations and evidential relationships remained vulnerable to overstatement. Notably, accumulated knowledge enabled the construction of longer mechanistic chains (12 and 13 edges) without increasing the number of edges verified against their own source (one per chain), although a few agent-proposed edges were supported indirectly by other verified evidence.

These findings demonstrate the feasibility of integrating scientific agents into an auditable discovery workflow while exposing a limitation of persistent agentic knowledge: structural coherence and knowledge accumulation do not necessarily imply stronger evidential support. The system generates falsifiable therapeutic hypotheses but establishes neither validated causal mechanisms nor therapeutic efficacy.

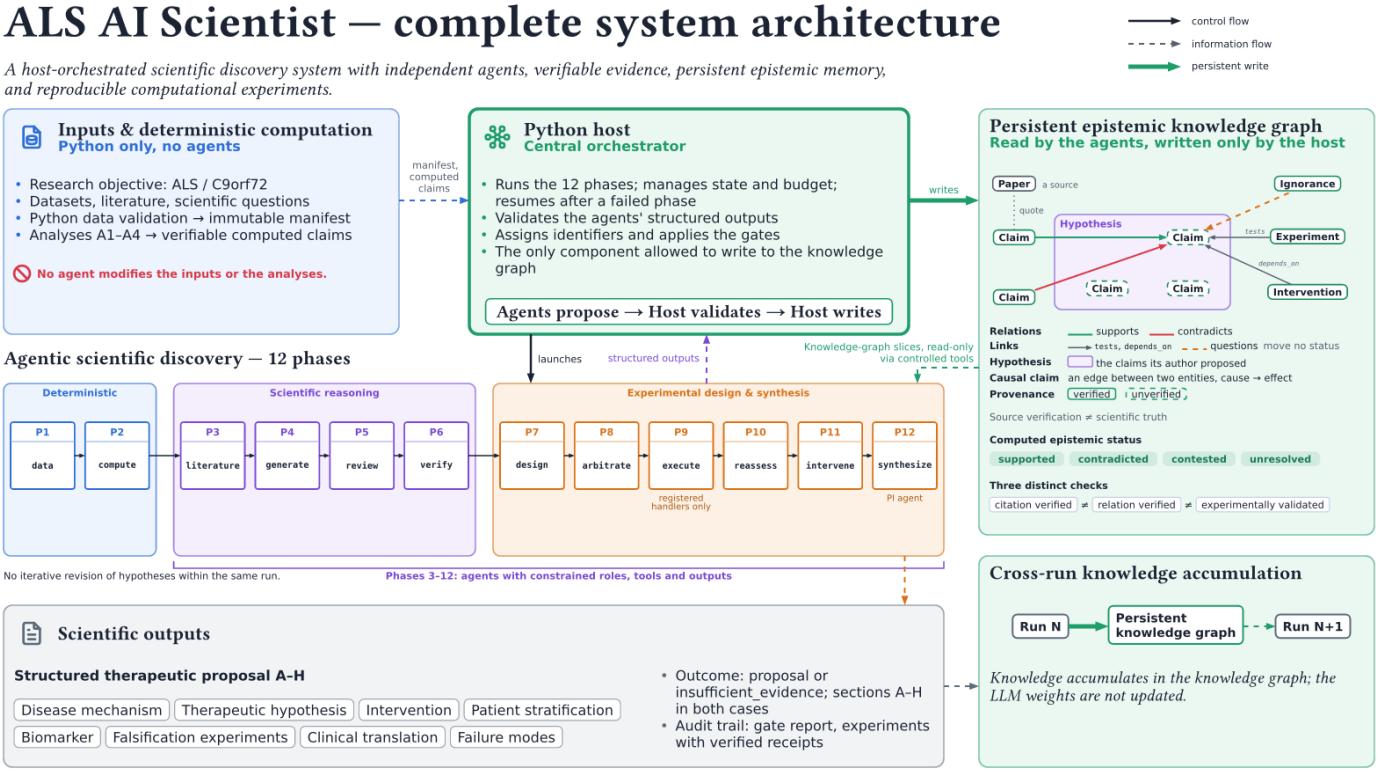


Figure 1. Architecture of the system. The Python host runs the twelve phases, validates the agents' structured outputs and is the only component that writes the knowledge graph; agents read it through read-only tools.

1. Introduction

Amyotrophic lateral sclerosis (ALS) remains a major therapeutic challenge, in part because its clinical phenotype can emerge from heterogeneous and only partially understood molecular and immune mechanisms. A recent study by Michaelis *et al.* identified C9orf72 as a target of ALS-associated T-cell autoreactivity, with predominantly $CD4^+$ responses, phenotype differences associated with C9orf72 mutation status, and an association between IL-10-mediated responses and longer predicted survival. These observations motivate the hypothesis that loss of peripheral tolerance to C9orf72 may participate in ALS pathophysiology and provide a concrete starting point for investigating antigen-specific therapeutic strategies [15].

At the same time, the rapidly expanding biomedical literature makes mechanistic hypothesis generation increasingly difficult to perform exhaustively. Literature-Based Discovery (LBD) was developed precisely to expose previously unconnected knowledge across scientific literatures, but classical approaches remain limited by scalability, dependence on structured resources, and substantial expert curation [9]. Large language models (LLMs) extend this paradigm by supporting literature synthesis, structured reasoning, and hypothesis generation; recent scientific agents have demonstrated strong performance in literature retrieval, synthesis, and contradiction detection [20]. Nevertheless, generation alone is insufficient for autonomous science. Machine-generated hypotheses remain difficult to assess with respect to novelty, plausibility, feasibility, and falsifiability [1], while intrinsic LLM self-correction can fail or even degrade reasoning in the absence of external feedback [7].

This work therefore addresses a stronger problem than scientific question answering: whether an agentic system can progress from an initial biomedical observation through mechanism construction, competing

hypotheses, computational analysis, intervention selection, adversarial criticism, and ultimately falsifiable experimental proposals. The aim of this work is an end-to-end *ALS AI Scientist* capable of converting the C9orf72 autoimmune finding into a traceable therapeutic research programme rather than merely summarising the literature.

We implement this objective as a reproducible multi-agent discovery workflow centred on a persistent knowledge base of atomic claims. Literature, hypothesis-generation, adversarial-review, verification, experimental-design, intervention and principal-investigator agents are separated by role. Agents propose information but do not directly modify epistemic state: a deterministic host validates structured records and provenance before committing them to an append-only knowledge base. Hypotheses are decomposed into claims and linked to supporting, contradicting, experimental and intervention records, enabling the final proposal to be traced back from conclusions to individual evidence statements. This explicitly separates provenance, that is whether evidence can be traced to its source, from epistemic support, that is whether independently checked evidence supports or contradicts a claim.

The contribution of this work is consequently both scientific and architectural. Scientifically, we investigate mechanistic and therapeutic hypotheses connecting C9orf72-associated immunity to ALS progression. Architecturally, we develop an auditable agentic discovery system in which generation, criticism, verification and falsification are distinct operations, computational analyses produce reproducible evidence, and the final synthesis may explicitly return *insufficient evidence* rather than manufacture a therapeutic conclusion. The resulting platform is intended as a test of whether multi-agent AI can support a defensible transition along the following chain: observation, mechanism, hypothesis, intervention and falsification experiment.

2. Background and Related Work

2.1 The Starting Observation

Michaelis et al. [15] investigated T-cell autoreactivity against C9orf72 peptides in 40 patients with amyotrophic lateral sclerosis (ALS) and 28 healthy controls. They observed increased C9orf72-specific responses in ALS, particularly among carriers of the *C9ORF72* repeat expansion, with elevated IL-5 and IL-10 responses. Notably, IL-10-mediated responses were associated with longer predicted survival, suggesting a potential role for regulatory immune mechanisms in disease progression. Building on these findings, this work implements an autonomous multi-agent system aimed at generating mechanistically explicit and experimentally falsifiable therapeutic hypotheses for ALS. The following sections review the scientific discovery architectures and evidence-grounding strategies relevant to this objective.

2.2 AI Systems for Scientific Discovery

Automated scientific discovery encompasses computational approaches to knowledge synthesis, hypothesis generation, experimental design, and validation. Early *Literature-Based Discovery* (LBD), pioneered by Swanson, sought previously unexplored connections between scientific concepts through intermediate associations. Large Language Models (LLMs) have since extended this idea from linking concepts to automating several stages of research: increasingly autonomous AI Scientists now coordinate multiple stages of scientific research, such as Sakana's AI Scientist [11] and Google's Co-Scientist [6]. The breadth of the resulting field is mapped by a survey of hypothesis generation with language models [1], which organises it into six families: literature-based discovery, human-centric methods, text mining, supervised learning, graph-based methods and LLM-driven approaches. It names hallucination, interpretability, bias, cost and domain adaptation among the open challenges and points to provenance tracking as a direction for future work. It classifies approaches by technique and reports no quantitative comparison between them.

We believe that a further distinction by architecture helps to understand how the stages of these systems interact. AI Scientists differ in how scientific tasks are organized, how candidate hypotheses are explored, and how feedback informs subsequent decisions. Four representative architectural patterns can be distinguished, although they are not mutually exclusive.

- **Sequential pipelines.** In its focused mode, which starts from human-provided code templates, The AI Scientist [11] organizes scientific discovery into successive stages, including idea generation, experimental implementation, execution, manuscript preparation, and automated review. This decomposition supports task specialization and reproducibility, although predominantly sequential execution may constrain the exploration of alternative research directions.
- **Iterative refinement.** Self-Refine [13] introduces a generation-feedback-refinement loop using a single LLM, while Reflexion [19] incorporates episodic memory to preserve feedback across attempts. Both approaches enable iterative improvement without updating model parameters. However, their effectiveness depends on feedback reliability: Huang et al. [7] demonstrated that intrinsic self-correction can degrade reasoning accuracy when reliable external feedback is unavailable.
- **Tree-search architectures.** The template-free mode of the same system, first released as AI Scientist-v2 [24], employs progressive agentic tree search to explore and evaluate alternative experimental trajectories [11]. Unlike predominantly sequential approaches, tree search allows multiple research directions to be considered before selecting promising branches, at the cost of additional computation and evaluation.
- **Population-based multi-agent systems.** Co-Scientist [6] combines specialized agents for generation, reflection, ranking, evolution, proximity assessment, and meta-review. Candidate hypotheses undergo iterative criticism, pairwise comparison, and refinement using external scientific evidence. This architecture supports hypothesis diversity and systematic comparison, although its effectiveness depends on the reliability of automated evaluation and ranking.

Beyond workflow topology, scientific discovery systems differ in how they interact with external tools, computational resources, and experimental environments.

- **Experiment-driven multi-agent workflows.** Robin [5] coordinates specialized literature-search and data-analysis agents within an iterative scientific discovery process. Results obtained from laboratory experiments are analyzed and used to inform subsequent hypotheses, connecting computational reasoning with empirical feedback while retaining human involvement in experimental execution.
- **Dynamic tool-composition architectures.** Biomni [8] combines a general-purpose planning agent with a unified biomedical environment containing computational tools, software, and databases. Through tool retrieval and code-based execution, the agent dynamically constructs and revises analytical workflows according to the scientific task, rather than relying exclusively on predefined pipelines.

Finally, workflow architecture itself influences performance. Kim et al. [10] investigated 180 configurations across five agent architectures, one single-agent and four multi-agent, finding that performance depends strongly on task characteristics and coordination strategy rather than simply increasing the number of agents. This highlights the importance of selecting an appropriate orchestration strategy for the scientific task.

Overall, existing research demonstrates multiple approaches to hypothesis exploration, iterative criticism, scientific tool integration, and evidence retrieval. Integrating these capabilities into a coherent scientific discovery workflow requires not only coordinating specialized agents, but also maintaining reliable evidence and evaluation mechanisms throughout the hypothesis generation and refinement process.

2.3 Scientific Reasoning and Hypothesis Evaluation

Scientific reasoning in AI Scientists involves generating, comparing, criticizing, and revising hypotheses using available scientific knowledge and evidence. Unlike conventional text generation, scientific hypothesis evaluation must consider not only coherence and plausibility, but also empirical support, novelty, feasibility, and falsifiability [1]. Several approaches have been proposed to support this process.

Iterative refinement methods, such as Self-Refine [13] and Reflexion [19], improve candidate outputs through feedback and repeated revision. However, Huang et al. [7] demonstrated that intrinsic self-correction may degrade reasoning performance when reliable external feedback is unavailable. This limitation is particularly relevant to scientific discovery, where the correctness of a novel hypothesis is generally unknown.

Adversarial and collaborative reasoning addresses this limitation by introducing specialized agents that challenge proposed hypotheses. Co-Scientist [6], for example, employs reflection, ranking, and evolutionary refinement to compare competing research directions. Nevertheless, agreement between multiple LLM agents does not independently establish scientific validity. Automated criticism can identify inconsistencies and weaknesses, but its conclusions remain dependent on the information and evaluation criteria available to the agents.

Evidence-grounded evaluation complements these strategies by connecting scientific claims to external sources. PaperQA2 [20] demonstrates agentic literature retrieval, evidence synthesis, citation traversal, and contradiction detection. Such mechanisms can support the verification of factual premises, although retrieving a relevant publication does not guarantee that it supports the claim attributed to it. Citation hallucinations and unsupported interpretations therefore remain important risks, particularly when generated claims are reused across successive reasoning stages.

Finally, scientific hypotheses must be distinguished from established findings. In particular, causal statements extracted from literature do not necessarily represent independently validated causal effects. For biomedical discovery, hypotheses should consequently be assessed through explicit supporting and contradicting evidence and translated into experiments capable of challenging their underlying mechanisms.

Together, these considerations motivate scientific discovery workflows that combine hypothesis exploration, independent criticism, evidence verification, and falsification-oriented experimental reasoning, rather than relying exclusively on agent consensus or iterative self-evaluation.

2.4 Scientific Evidence Grounding and Knowledge Representation

Scientific discovery systems require access to reliable external knowledge to support hypothesis generation and evaluation. Retrieval-Augmented Generation (RAG) addresses this requirement by incorporating retrieved documents into the generation process. However, conventional RAG predominantly relies on semantic similarity, which does not necessarily capture the relationships between scientific findings, their underlying assumptions, and their supporting evidence.

Agentic scientific retrieval. PaperQA2 [20] extends conventional RAG through iterative literature search, evidence ranking, citation traversal, and contradiction detection. Paper2Agent [14] takes a complementary approach by transforming scientific publications, supplementary materials, and computational resources into executable interfaces accessible to AI agents. These approaches improve access to scientific knowledge and computational methods, but access to a source does not, by itself, establish the validity of claims derived from it.

Graph-based knowledge representation. Knowledge graphs organize information through explicit relationships between entities or propositions, enabling structured retrieval beyond isolated text passages. CausalRAG

[21] incorporates LLM-inferred causal paths into retrieval. CausalRAG2 [22] extends this approach through a hierarchical knowledge graph connected by causal gates, which facilitate retrieval across otherwise isolated modules. Importantly, the authors define causality in terms of logical dependencies and event sequences expressed in text, rather than statistical causal discovery. These methods therefore address evidence retrieval and contextual organization, rather than the identification of causal effects through interventions.

Evidence provenance and verification. A major challenge in LLM-based scientific discovery is distinguishing retrieved information from verified scientific evidence. Citation hallucinations, unsupported interpretations, and contradictory findings can propagate through successive reasoning stages, potentially influencing hypothesis selection and experimental design. Furthermore, agreement among multiple agents does not provide independent evidence of factual correctness. Scientific knowledge representations therefore benefit from explicit provenance, claim-level attribution, and mechanisms for distinguishing supporting, contradicting, and unresolved evidence.

Together, these approaches illustrate the importance of combining flexible scientific information retrieval with structured and verifiable knowledge representations. For autonomous scientific discovery, the central challenge is not merely retrieving relevant information, but maintaining traceable relationships between scientific claims, their supporting evidence, and the hypotheses derived from them.

2.5 Positioning: This System vs. Related Systems

This system is a multi-agent pipeline for therapeutic hypothesis generation in ALS, grounded in the C9orf72 immune findings of Michaelis et al. [15]. It covers literature acquisition, a persistent knowledge base of claims with provenance and supporting or contradicting relations, hypothesis generation, adversarial review, comparison of hypotheses and the proposal of experiments that could falsify them. Relations between hypotheses, evidence, objections and revisions are stored explicitly, and a claim's status follows defined evidence validation rules, not agent consensus alone. Although the runs reported in Section 6 provide neither biological validation nor evidence of therapeutic efficacy, they show that such a workflow can be integrated end to end.

The reliability of the generated claims and the behaviour of the verification steps are examined in Sections 5 to 7; no ablation isolates each component, and the proposed experiments were not assessed by domain experts. The work is positioned as auditable infrastructure for AI assisted discovery, built on traceable evidence, adversarial refinement and actionable experiments. Table 1 compares it with the systems above on the properties that this design chooses to enforce. The comparison is not a benchmark and does not rank the systems: the column for this work describes behaviour implemented in its code, whereas every other column records what the cited paper reports, so a difference between columns can reflect how much a paper documents rather than what its system does.

Causal graphs over text: retrieval and audit. On how a graph over text decides what counts as a causal link, the closest comparison is the CausalRAG line [21, 22], which pursues an objective other than claim-level audit, namely retrieval-augmented question answering. CausalRAG2 adds **causal gates** between modules of a Leiden partition, where they "use the term *causal* to denote explicit logical dependencies and event sequences described in the text, rather than statistical causal discovery from observational data" [22]. A gate is an undirected, binary LLM judgement between two modules that selects context for retrieval; its contribution is measured by an ablation fixing the gate count and varying the selection rule (causal 31.62 F1, semantic 24.87, random 23.86) [22, Table 6]. In our system, edges are typed relations between identified claims, written by the host and never by an agent, and status is computed from **verified relations only**.

The two designs therefore optimise different properties. CausalRAG2 admits a link when an LLM judges two modules to depend on each other, which suits retrieval, where links are evaluated through the quality of

the answers. In our system, a link is a causal claim only when it names a cause, an effect, a context and a mechanism, and it counts as evidence only when its quotation is found in the stored source and a verification agent confirms the relation, so each step leaves a record that can be re-checked, at the cost of recall (§4.2). Such verification establishes that the documentary support is admissible, not that the relation is causally identified. Neither approach shows that a link is biologically true, and CausalRAG2 does not claim to. In our system, a quotation can be found in its source and still be stretched beyond what it says (§3.3). The verifier catches such a stretch, including an association read as causation, when the claim serves as evidence for another. A causal claim that only enters a mechanism chain, however, is checked for its quotation and nothing else, so an association stated as causation can pass unnoticed.

	Co-Scientist Gottweis et al. 2026	**Sakana AI Scientist** Lu et al. 2026	**Robin** Ghareeb et al. 2026	**Biomni** Huang et al. 2026	**This work**
Hypothesis generation	Population of specialized agents: generation, reflection, ranking, evolution	Idea generation stage of a sequential pipeline	Yes, revised from analysed experimental results	Task driven planning agent	Three independent authors, one per perspective
Computational execution and tool use	Not reported	Yes: implements and runs its own experiments, then writes the manuscript	Data analysis agents; laboratory experiments executed by humans	Yes: retrieves and composes biomedical tools, databases and code	Registered deterministic analyses and experiment handlers only
Falsification framework	Implicit (hypothesis refinement)	Not reported	Not reported	Not reported	Explicit, experiment level falsification criteria
Evidence state / gating	Not reported	Not reported	Not reported	Not reported	Host computed from verified relations
Adversarial critique	Implicit (reflection and ranking)	Implicit (automated review)	Not reported	Not reported	Blinded reviewers, arbitration may reject
Insufficient evidence outcome	Unclear	No (converges on a paper)	No (produces a proposal)	No (converges on an analysis)	Explicit insufficient evidence outcome
Reproducibility mechanisms	Not reported	Version control, GitHub	Checkpointing	Not reported	Archived prompts, run artifacts, SHA-256 digests
Source / citation verification	Not reported	Implicit	Implicit (literature search agents)	Not reported	Quote check at retrieval plus independent relation verification

Table 1. Comparison with autonomous scientific discovery systems. Each cell is a statement about what the cited paper reports, never about the system itself: *Yes*, the source explicitly supports the capability; *No*, the source clearly shows it is absent; *Not reported*, the paper does not describe it; *Unclear*, partial or ambiguous; *Implicit*, the behaviour exists but is not formalized. Enforcing explicit, experiment-level falsification criteria is motivated by work on falsifiable research ideation and on adversarial experiments in agentic science [4, 23].

2.6 Known Gaps and Limitations in Related Work

1. **Falsification-first implementation:** Fa & Culjak [4] and Wang et al. [23] converge on falsification-first standards, but full implementations in biomedical discovery are nascent.

2. **Adversarial critique in ALS:** we found no published work applying adversarial hypothesis evaluation to ALS mechanistic hypotheses.
3. **Reproducibility and causal inference:** we found no systematic reproducibility and comparison framework for multi-agent workflows, and no published implementation of causal inference with falsification.
4. **Evaluation:** most systems are validated on computational tasks or simulation, with limited validation on open-ended biomedical questions. Luo et al. [12] identify four failure modes, namely inappropriate benchmarking, data leakage, metric misuse and post-hoc selection bias, and argue that journals and conferences should audit full workflow artifacts rather than manuscripts alone; our provenance infrastructure (Section 4) is a partial response.

Items 2 and 3 record what the searches behind this section did not find. No systematic review was conducted, so a proper assessment of the novelty of this work is left to future work.

These gaps motivate Sections 3 to 5.

3. Methods

3.1 Problem Formulation

We formulate autonomous scientific discovery as an evidence-constrained process in which specialised language-model agents generate mechanistic hypotheses, challenge their assumptions, and propose falsifiable investigations. The system addresses amyotrophic lateral sclerosis (ALS), starting from the C9orf72-associated immune response reported by Michaelis et al. [15]. Its objective is to identify plausible therapeutic research directions without supplying a predefined intervention. The intended output is a traceable research proposal, not a demonstrated treatment or an experimentally established causal model.

Each run takes the research objective, the declared datasets, the scientific literature and the knowledge base accumulated by earlier runs, and returns the extended knowledge base together with either a structured therapeutic research proposal or an explicit `insufficient_evidence` outcome.

Three principles govern the method: **traceability**, linking assertions to identifiable records and evidence; **separation of responsibilities**, preventing the agents that make claims from unilaterally verifying them; and **falsifiability**, requiring candidate experiments to state observations that would challenge specific claims. In particular, an assertion extracted from a publication is not treated as a demonstrated causal effect merely because the publication contains causal language.

The principles are enforced by **gates**. In this report a gate is a deterministic check that the host, never an agent, runs on an agent's output before admitting it: a *blocking* gate rejects the record or stops the run, a *diagnostic* gate admits the output and records a warning. A *structural* gate checks the form of an output, for instance that identifiers exist, that cited causal claims connect or that an experiment has a receipt, never whether its content is true.

3.2 Multi-Agent Discovery Workflow

The architecture divides discovery into twelve ordered phases spanning evidence acquisition, hypothesis generation, adversarial assessment, experimental planning, computational reassessment, and therapeutic synthesis. Two phases (`data` and `compute`) execute deterministic procedures; the remaining ten use role-specific LLM agents. Agentic literature research and adversarial hypothesis assessment build on the broader literature on scientific retrieval and hypothesis generation [1, 20]. We do not assume that self-criticism alone

improves factual accuracy: empirical results on intrinsic LLM self-correction motivate externally grounded verification [7].

The phases form six stages. *Evidence acquisition* (`data`, `compute`, `literature`) inventories the datasets, computes auditable results and extracts sourced claims. *Hypothesis construction* (`generate`) proposes independent hypotheses from a shared causal prior. *Critical assessment* (`review`, `verify`) seeks counter-evidence and judges the proposed evidential relations. *Experimental planning* (`design`, `arbitrate`) specifies falsification tests and decides which to pursue. *Computational reassessment* (`execute`, `reassess`) runs the available registered analyses and interprets their results. *Therapeutic synthesis* (`intervene`, `synthesize`) proposes candidate interventions and writes the final research proposal.

The orchestrator supplies each agent with a task-specific subset of the knowledge base and an explicitly restricted tool set. Literature agents independently gather evidence; hypothesis authors work from the same prior without viewing competing authors' proposals; reviewers are blinded to author identity and declared confidence. The resulting agent outputs are proposals for structured records, never direct modifications of the knowledge base. A single host component validates and commits admissible records.

Successive runs inherit the accumulated knowledge base. This is **persistent evidential memory**, not model training: no model weights are updated, and hypotheses are not repeatedly rewritten by an author-reviewer loop within a run. Evidence added during later runs can affect the evaluation of existing claims.

3.3 Scientific Knowledge Representation and Mechanism Construction

A graph of atomic scientific records. The method stores scientific information as **atomic claims**: typed records, each with its own identifier and sources. This follows the premise of literature-based discovery: useful hypotheses can arise by connecting findings distributed across otherwise separate sources [9]. Unlike embedding-based GraphRAG or CausalRAG architectures [21], the implementation uses symbolic retrieval over typed records, with no vector embeddings or semantic nearest-neighbour search.

We distinguish two graph representations. The **knowledge graph**, $\mathcal{K} = (V, E)$, links typed records: claims, entities, hypotheses, ignorance records, experiments, decisions, interventions, and provenance events. Its relations include `supports`, `contradicts`, `questions`, `tests`, `part_of`, `derived_from`, `depends_on`, and `supersedes`. A hypothesis is associated with its constituent assertions through `part_of` edges, so those assertions remain individually inspectable. An ignorance record represents a question or missing evidence; it is not counted as evidence against a hypothesis.

The **mechanism graph** is a derived view over *causal claims*. A claim qualifies as causal when it specifies a cause entity, an effect entity, a context, and a proposed mechanism. Each such claim contributes a directed edge from its cause to its effect, annotated with its record identifier, provenance and epistemic assessment. Several claims can assert relationships between the same entities, making this a directed multigraph. Unlike a structural causal model, this graph contains only relations asserted in text or by agent reasoning: the system does not estimate causal effects, identify interventions, or adjust for confounding when constructing it.

Provenance, evidence verification and epistemic status. Support for a claim is checked at three distinct levels, and the system automates only the first two:

1. **Syntactic provenance**: the declared source exists and contains what is attributed to it. For a literature claim, a deterministic check finds the verbatim quotation in the stored paper text; for a computational claim, an attributable analysis artifact exists. A pass yields *verified provenance*.
2. **Semantic entailment**: the source actually supports the claim, or the relation, attributed to it. An independent verification agent judges it, as described below, and its judgements have not been calibrated against a human expert.

3. **Empirical validation**: the claim holds in the world. No component of the system establishes it; it requires experiments that the system can only propose.

In this system each level is a precondition for the next by rule, not by an epistemological principle: a relation counts towards a claim's status only if the evidence claim it comes from has verified provenance (`knowledge_base/status.py::is_active`), and *supported* is computed only from relations confirmed at level 2. A claim can be true without any stored source. No level implies the next:

$$\text{Source verified} \not\Rightarrow \text{Claim supported} \not\Rightarrow \text{Claim true} \tag{1}$$

Throughout this report, a *verified claim* has passed level 1, a *verified relation* has been confirmed by the verification agent at level 2, and *supported* is the status computed from such relations; the workflow does not independently establish level-3 validation, which would require experimental evidence it does not produce.

Claims written by a computation, in the `compute` phase or from a receipted `execute` result, pass level 1 because re-running the analysis on the declared inputs reproduces them. No relation is written from them: within a run, a computational result changes no claim's status, and the `reassess` verdict (§3.5) is archived as a report. A later run can assert relations from these claims, which then pass through `verify` like any other.

Agents may propose supporting or contradicting relations between claims, but those assertions initially remain unverified. A separate verification phase examines the source quotation and the proposed relation. Only confirmed relationships satisfying the verification conditions can affect epistemic status. If a quotation supports only a narrower non-causal statement, the implementation can create a separate, narrower claim while preserving the original; this automatic narrowing procedure is not applied to causal claims. An unsupported contradiction can be retained as a question, which does not count as evidence against the claim.

For a claim c, let $S(c)$ and $C(c)$ be its active incoming `supports` and `contradicts` relations. A relation is active only when it has verified provenance, originates from a verified and current source claim, and is not excluded by the record rules. The claim's status is computed from these relations:

$$\text{status}(c) = \begin{cases} \text{contested}, & S(c) \neq \varnothing \ \wedge \ C(c) \neq \varnothing, \\ \text{supported}, & S(c) \neq \varnothing \ \wedge \ C(c) = \varnothing, \\ \text{contradicted}, & S(c) = \varnothing \ \wedge \ C(c) \neq \varnothing, \\ \text{unresolved}, & \text{otherwise.} \end{cases} \tag{2}$$

The labels describe the currently recorded relations among claims, not probabilities of truth. A claim may therefore have verified provenance yet remain unresolved, or have unverified provenance while receiving support from another verified claim. No corresponding aggregate epistemic-status rule is implemented for whole hypotheses: arbitration and final synthesis reason over their constituent claims and experimental evidence without assigning hypotheses a formally computed status, a pairwise tournament ranking, or a calibrated posterior probability.

Prior causal graph and mechanism backbone. At the end of the literature phase, the host constructs a frozen prior causal graph by selecting causal claims from the current run that originate in literature and have quote-verified provenance. This graph gives the independent hypothesis authors a common scaffold. It is distinct from the mechanism reconstructed for the final proposal.

For synthesis, the system queries the accumulated mechanism graph and selects a backbone: a longest admissible directed *simple path* of causal claims. Consecutive claims must share the same intermediate entity,

no entity may recur, the default minimum path length is two edges, and the selected path must contain at least one edge with verified provenance. Among admissible paths, selection maximises the number of edges; it does not optimise evidence strength or biological plausibility. A surrounding neighbourhood additionally exposes non-backbone causal assertions with unverified provenance that touch entities on the path.

This design turns an otherwise disconnected list of causal assertions into a readable mechanistic chain while preserving access to alternative and uncertain links. Its limitations are material: a long chain may contain mostly unverified assertions; exact entity naming can break connections; and the backbone may join claims originating from different hypotheses or earlier runs. Accordingly, we describe it as a chain with at least one quote-checked edge, never as a verified causal pathway. The section-A gate of the runs reported here admitted such a chain whatever the share of unchecked edges; it was subsequently restricted to a backbone of checked edges plus at most one unchecked edge adjacent to it, after this series measured the consequence. For example, the archived run `pl-prod-016` yields a six-edge backbone with only two provenance-verified edges and ultimately reports `insufficient_evidence`.

3.4 Independent Generation, Adversarial Review and Verification

The `generate` phase assigns independent author agents (or simply authors) a shared scientific objective, a bounded knowledge-base view, and the prior causal graph. Authors propose explanatory mechanisms, decompose them into constituent claims, link them to sources or explicitly mark them as inference, and record unresolved questions. The prior gives a common vocabulary while directing attention towards missing upstream explanations, unexplained effects, and unresolved or contested causal links. This procedure encourages multiple candidate mechanisms without asserting that their novelty is quantitatively established.

Each candidate is then examined by at least two independent, blinded reviewers. Their purpose is not simply to improve a draft but to identify unsupported premises, alternative explanations, adverse evidence, and experimentally discriminating questions. A proposed objection is not equivalent to confirmed counter-evidence. Reviewers' relations enter the shared store as unverified inferences, and an independent `verify` phase evaluates eligible support and contradiction links using their quoted evidence. Confirmation can produce a verified relation; rejected or insufficient judgments do not change the target claim's epistemic status.

These boundaries are important in light of the limitations of intrinsic self-correction [7]. The system separates generation, criticism, and admissibility of evidential relations. It does not claim that independent LLM reviewers are statistically independent observations, nor that agreement among them proves a biomedical claim. Likewise, there is no implemented within-run hypothesis-rewriting loop: the principal mechanisms of improvement are exposure to alternative evidence, explicit questions, controlled verification, and accumulated knowledge across runs.

3.5 Evidence-Grounded Experimental Arbitration

Experimental design is tied to individual claims. A candidate experiment must specify at least two possible outcomes, of which at least one would contradict a claim it tests. This is a schema-level falsifiability requirement, not a guarantee of sufficient statistical power or decisive biological identification. Experiments are linked to the claims they test, allowing the system to retain the distinction between an attractive hypothesis and a feasible investigation.

The `arbitrate` phase examines the hypothesis neighbourhood, evidence and available feasibility information, returning one of `buy`, `rejected`, or `stop`. Rejection based on contradiction is constrained to verified evidence against an indispensable claim. These decisions are agent judgments governed by structured input and

validation rules, not solutions of a formal optimal experimental-design objective. Nor does a decision to pursue an experiment validate the hypothesis.

The `execute` phase runs only computational experiments supported by registered analysis handlers. A proposed or selected experiment is not reported as performed unless an execution receipt verifies its result. The receipt is a record written by the host after a handler runs: it re-hashes the declared inputs and the output artifacts, binds them to the analysis identifier and the digest of its result, and is itself sealed by a digest, so that anyone can recompute the result and compare. A *receipted computation* is a result that carries such a receipt. The subsequent `reassess` phase interprets available results, but does not automatically write an accept/reject status onto the hypothesis record. Thus, proposed tests, computationally executed tests, and biological validation remain distinct categories.

3.6 Therapeutic Synthesis and Output Constraints

After experimental arbitration and any supported computational analyses, specialised agents propose candidate interventions attached to the claims on which they depend. Intervention records require explicit off-target risks, safety concerns, a patient-stratification rationale, and a candidate biomarker. The principal-investigator (PI) agent then integrates the available mechanisms, hypotheses, experiments, decisions and intervention records into an eight-part therapeutic research proposal.

The proposal has eight sections: (A) a disease mechanism, the asserted causal chain from C9orf72-associated processes to ALS; (B) a testable therapeutic hypothesis; (C) a candidate intervention and its target; (D) the patient subgroup most likely to benefit; (E) a biomarker of biological response; (F) validation experiments whose outcomes could challenge the hypothesis; (G) a preclinical-to-clinical translation path; and (H) at least five plausible reasons the intervention might fail.

Each section cites the identifiers of the knowledge-base records it rests on. A proposal gate checks that citations are syntactically valid and refer to existing records, and blocks unsupported claims that an experiment was performed or runnable without an appropriate verified receipt. Other quality checks raise warnings and do not block the output: whether the mechanism references connected causal claims, whether required intervention records are cited, and whether cited claims are contested or overstated. Thus, passing the gate establishes particular structural and traceability conditions, not scientific correctness.

The PI may output `insufficient_evidence`, preserving a defensible distinction between generating plausible hypotheses and recommending a therapy. The scientific endpoint is a reproducible chain from sources or computations to atomic claims, proposed mechanisms, falsification experiments, and an auditable research proposal. Independent biological validation and demonstration of clinical benefit lie outside the evidential guarantees of this workflow.

4. Implementation

The discovery platform is implemented as a configurable Python orchestration pipeline operating over Claude Code agent workflows and a persistent, typed knowledge base. The implementation separates three execution boundaries: language-model agents propose structured outputs, the Python host validates and commits records, and deterministic analysis handlers compute results from declared datasets. This separation is central to the system's auditability: neither an agent response nor a successfully completed workflow phase is, by itself, accepted as verified scientific evidence.

4.1 Software Architecture and Agent Execution

The main entry point is the Python orchestrator (`pipeline/orchestrator.py`), which executes twelve ordered phases: `data`, `compute`, `literature`, `generate`, `review`, `verify`, `design`, `arbitrate`, `execute`, `reassess`, `intervene`, and `synthesize`. The first two are ordinary software procedures; the remaining ten use agent workflows. The scientific objective, dataset sources, phase settings, and resource limits are specified in `configs/pipeline.yaml`. The configured objective is injected into the relevant role prompts, rather than supplied interactively during each run.

Agent roles are defined as Markdown specifications (`agents/<phase>.md`). The host launches phase-specific Claude Code sessions through `claude -p`, using *dynamic workflows*, a Claude Code feature [2], to coordinate specialized agents within each phase. Workflow sources are implemented in JavaScript and compiled using `workflows/build_workflow.mjs`. Thus, JavaScript governs intra-phase agent coordination, while Python controls cross-phase sequencing, validation, persistent state, retries, and run accounting. A phase with no eligible work can be skipped. These components are not independent chatbot conversations: they exchange information via validated phase outputs and selected knowledge-base views.

The number and independence of agents depend on the phase. Literature questions and hypothesis-generation perspectives are assigned to separate agents; hypotheses are reviewed by multiple blinded reviewers; verification agents inspect batches of proposed evidence relations; synthesis is performed by a single principal-investigator agent. The host limits each phase's available tools. Agents may access read-only knowledge-base tools and, where configured, literature or web tools, but cannot directly write the shared knowledge base or execute arbitrary shell commands. The `execute` phase uses registered computational handlers rather than executing free-form code produced by an agent.

Each agent phase is governed by a typed output contract defined with Pydantic models in `pipeline/proposals.py`. Schemas supplied to the workflow are generated from these models, reducing divergence between agent instructions and host-side validation. Unknown fields are rejected. Invalid items are handled at an appropriate granularity: the host can normalize permissible provenance inconsistencies, omit invalid claims or relations with recorded reasons, or reject an output that cannot be parsed. This distinction prevents a single malformed record from silently contaminating the store or unnecessarily invalidating all otherwise usable output.

4.2 Knowledge-Base Storage and Graph Construction

The knowledge base is implemented as an append-only collection of JSON Lines files, separated by record type. Its nine typed record categories are claims, relations, entities, hypotheses, ignorance records, events, experiments, decisions, and interventions. A single-writer lock protects mutations, and agents can interact with the store only through read-only retrieval tools or structured proposals returned to the host. The host assigns persistent identifiers and resolves agent-local references when constructing records. Records are not overwritten to correct scientific statements; superseding records and event logs preserve their history.

Claims are deduplicated using a SHA-256 hash of normalized claim content and, when applicable, its causal fields. Source quotations are retained as provenance but do not determine claim identity. Re-encountering an existing claim produces a `seen_again` event rather than a second scientific assertion; references to the duplicate are redirected to the established identifier. Named entities are reused through conservative, rule-based string normalization. This avoids merging conceptually distinct biomedical entities on semantic similarity alone, although unrecognized synonyms can fragment causal paths.

A literature claim is eligible for `verified` provenance only when an exact quotation is found in the locally stored text of its declared source. This check establishes syntactic provenance only (§3.3). Computed claims

originate from registered analyses and refer to their input and output artifacts. The independent `verify` phase subsequently examines proposed `supports` and `contradicts` relationships for evidential faithfulness. Confirmed relationships are committed separately from the agent's initial, unverified inference. When a non-causal assertion overstates its quotation, a narrower claim may be recorded without deleting the original. Unsubstantiated objections can instead be retained as questions, without being counted as contradictory evidence.

Two graph views are derived from the same records. The **evidence graph** links claims, hypotheses, experiments and other records through typed relationships; its epistemic states are computed at read time from active, verified support and contradiction relations. The **mechanism graph** projects each current causal claim onto a directed edge between its referenced cause and effect entities. Causal assertions are distinguished by a complete causal field block (`cause_ref`, `effect_ref`, `context`, `mechanism`), not by a standalone causal type label. The graph is a multigraph of assertions, not a causal structure learned or identified from experimental data.

At the end of the literature phase, the host creates a versioned *prior causal graph* containing the run's literature-origin causal claims with verified quotation provenance. Hypothesis authors receive this fixed scaffold. Separately, a read-only mechanism-path tool derives the final candidate *backbone* from the accumulated graph: it enumerates directed simple paths and selects the longest path meeting the configured minimum length and containing at least one quote-verified edge, resolving ties by traversal order: the first longest path found wins, with the traversal starting from the lowest cause-entity identifier and following edges in claim-identifier order. A neighbouring view retains unverified causal edges incident on backbone entities. Critically, this selection maximizes path length, not evidential strength; a backbone can contain multiple unverified assertions, including claims inherited from earlier runs. It must therefore be presented as an inspectable mechanistic hypothesis chain, not as an experimentally established causal pathway. The tool selects the longest such path irrespective of the share of verified edges; the gate that admits the selected chain into a proposal was tightened after the runs reported here. The exhaustive path enumeration is also a potential scalability limitation for larger graphs.

Agent retrieval does not use embeddings or a vector database. The read-only MCP interface exposes record lookup, neighbourhood retrieval, deterministic keyword matching, and mechanism-graph/path queries. Retrieved contexts omit author identity and declared confidence where blinding is required, while including computed epistemic information. The resulting representation is directly inspectable, at the cost of limited recall for paraphrases or incompletely normalized entity names.

4.3 Pipeline State, Computational Execution and Final Artifacts

Every execution is assigned a run identifier and produces a structured archive under `experiments/<run-id>/results/`. The archive includes phase outputs and phase reports, a `state.json` checkpoint, agent interaction traces, and quote-check logs. The checkpoint records completed phases to support resumption; configuration changes on resume are recorded rather than silently ignored. The shared knowledge base normally persists across runs, whereas an explicitly selected empty knowledge-base directory enables an isolated execution. This distinction matters for evaluation because a later run may inherit evidence and unverified claims from earlier runs.

The `data` phase inventories configured sources, recording metadata and content hashes in a dataset manifest. Deterministic analyses produce verifiable summaries and computational claims. Agents receive dataset metadata without unrestricted access to arbitrary local files. For computational experiments, the `execute` phase can invoke only pre-registered handlers with an experiment identifier. An experiment is labelled *performed* only when the system has a verified execution receipt; otherwise, a selected experiment remains

a proposal. The subsequent `reassess` phase records an agent's interpretation of the results rather than modifying a formal hypothesis-level epistemic score, which is not implemented.

The terminal `synthesize` phase emits `final-proposal.json` and `final-proposal.md`, structured into eight sections (A to H). Scientific assertions are linked to knowledge-base identifiers, and the proposal gate checks citation syntax, existence of cited records, and claims that an experiment was executed without an admissible receipt. Additional checks are diagnostic rather than universally blocking: the connectedness of cited mechanism claims, the references supporting stratification and biomarkers, and the warnings for contested and unfaithful claims. The gate therefore supports structural traceability and honest reporting of execution, but cannot certify biological validity. The PI may return `insufficient_evidence` instead of presenting an inadequately supported intervention.

External biomedical services were accessed through deterministic host side clients or precomputed caches rather than directly by agents. In particular, binding predictions for human leukocyte antigen class II (HLA-II, the human major histocompatibility complex class II) molecules used a cached IEDB [3] NetMHCIIpan [17] resource consumed by deterministic analyses and experiment handlers. Europe PMC [18] and OpenAlex [16] supported literature retrieval, while other pharmacological endpoints were disabled in the reported production runs.

4.4 Reproducibility, Verification and Scope

Reproducibility rests on versioned configuration, declared dataset inputs and hashes, typed agent and record schemas, retained intermediate artifacts, phase transcripts, and resumable execution state. The repository provides a lower-cost smoke configuration (`configs/pipeline-smoke.yaml`) and automated test suites for the Python components and for the JavaScript workflows (`tests/workflow/*.test.js`, with mocked agents). Repeating calls to external LLMs or literature services may produce different scientific proposals despite identical software configuration. For exact scientific replay, archived model identifiers, prompts, configuration digests, inputs, agent outputs, and analysis receipts must be inspected together. Dataset and code-derived computations are more directly reproducible than open-ended generative decisions.

5. Experimental Setup

The experimental evaluation was designed to test the complete ALS AI Scientist as an end-to-end scientific discovery workflow, rather than evaluating isolated language-model responses. The scientific task was grounded in the C9orf72-associated immune mechanism described by Michaelis et al. [15]. The system was given a therapeutic objective without a predefined therapeutic solution: to identify an intervention capable of selectively modifying the C9orf72-associated immune mechanism in ALS while preserving systemic immune competence and protecting motor neurons.

The evaluation focused on two questions. First, we assessed whether independent executions of the same agentic workflow produced scientifically traceable and reproducible outputs when starting without accumulated knowledge. Second, we examined whether providing the system with an existing knowledge base changed the discovery process by allowing new agents to retrieve and reason over claims generated in previous runs.

This evaluation strategy treats scientific discovery as a multi-stage agentic process rather than as a single generation task, consistently with recent systems in which literature retrieval, evidence gathering, reasoning, and synthesis are exposed as separate agent actions [20].

5.1 Experimental conditions

Five complete production runs, `pl-prod-014` to `pl-prod-018`, were executed. All runs used the same scientific objective, pipeline configuration, agent definitions, and underlying LLM (`claude-sonnet-5`). The configuration hash was identical across runs. Differences between repository commits were restricted to documentation, reporting, and presentation code; the pipeline, agents, tools, schemas, analyses, knowledge-base implementation, and evaluation code were unchanged.

Two knowledge conditions were compared (Figure 2): three runs (014 to 016) started from an empty knowledge base, and two (017, 018) from one frozen accumulated snapshot of 1196 claims written by earlier runs (identical `kb-snapshot.sha256`), so the latter are two observations of a single initial state.

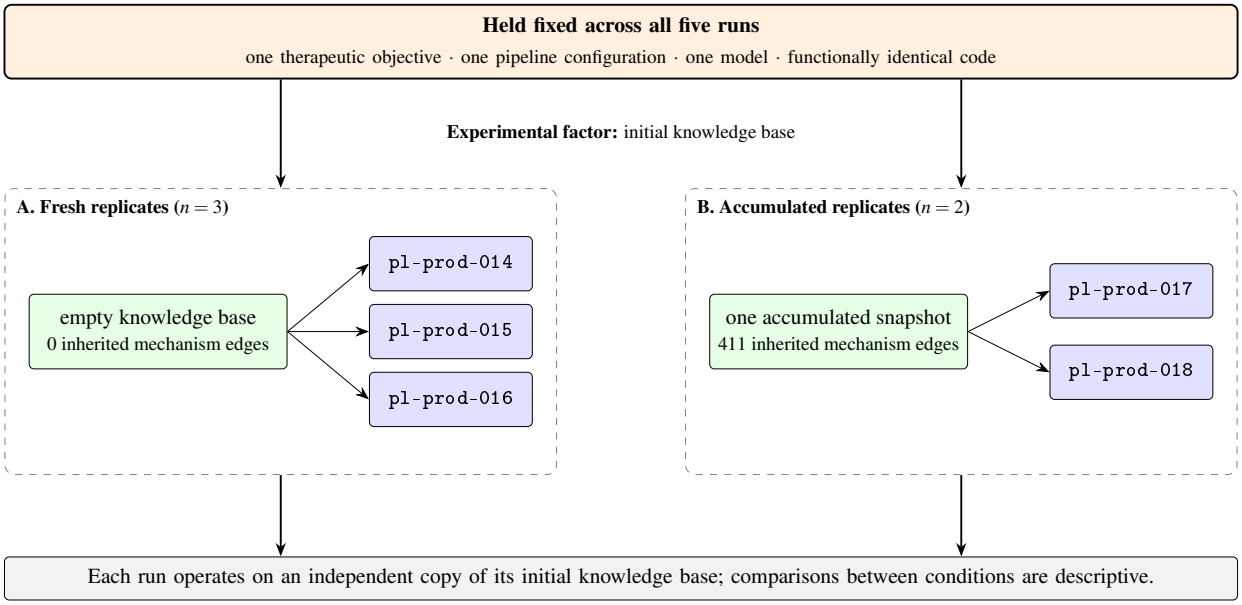


Figure 2. Experimental design. Five production runs share the therapeutic objective, the pipeline configuration (`config_hash a7e9eef5...`), the model (`claude-sonnet-5`) and functionally identical code; the experimental factor is the knowledge base at the start of a run. Each run operates on an independent copy of its initial knowledge base, and `verify` judges only the relations written by the current run. Results are reported per run: with $n = 3$ and $n = 2$, no inferential test compares the two conditions.

A shared full-text literature cache was retained across runs. Consequently, the knowledge-base state was reset for fresh runs, whereas previously downloaded papers could be reused. The experiment therefore isolates structured scientific memory only partially and should be interpreted as a knowledge-state comparison, rather than as a strict memory ablation.

5.2 Computational analyses and executable experiments

Every run included four deterministic analyses on the Michaelis et al. source data [15]:

1. immunodominance analysis;
2. epitope-clustering analysis;
3. GWAS-locus analysis;
4. HLA-binding analysis.

These analyses were executed independently of the generative stages and therefore provided a deterministic component within an otherwise stochastic agentic workflow.

Hypothesis-specific experiments were generated dynamically by the agents. To distinguish experimental design from executable computation, an experiment could only be executed automatically when a registered deterministic handler existed.

Two handlers were available:

- `hla_panel_compact_v1`;
- `ics_cytokine_skew_v1`, which re-analyses the four deposited intracellular cytokine staining screens of the source study [15].

Other proposed experiments, including wet-lab experiments, were retained as falsifiable experimental proposals but were not simulated or replaced by LLM judgement. This separation was intended to prevent linguistic assessment from being presented as experimental evidence. Table 2 lists the data resources, the deterministic analyses and the two registered handlers.

A separate novelty-assessment stage was executed after the twelve-phase core pipeline. Since this stage depended on an external literature API and could fail independently of the core workflow, novelty was treated separately from pipeline-completion status.

5.3 Reproducibility and recorded outputs

For every run, the system archived the configuration, model identifier, prompts and agent specifications, Git commit, knowledge-base state, phase reports, computational outputs, final proposal, tool-call counts, wall-clock time, and API cost. The unit of analysis was therefore one complete autonomous run, and every measurement reported below describes a whole run. Comparison across runs considered:

- final proposal outcome;
- generated hypotheses;
- proposed and executed experiments;
- verification verdicts;
- use of contradicted or contested evidence;
- causal-chain structure;
- citation checks;
- hypothesis diversity;
- computational cost;
- latency.

Deterministic computational analyses were additionally compared directly across runs to provide an internal reproducibility control.

5.4 Scope of the comparison

The experimental design does not support population-level statistical conclusions about the effect of accumulated knowledge. The fresh and accumulated conditions contain only three and two runs, respectively, and the two accumulated runs share the same initial knowledge snapshot. Accordingly, differences between conditions are interpreted as observed behavioural differences of the workflow, not as statistically established effects. The purpose of the experiment is instead to determine whether the system can repeatedly execute the complete scientific workflow, preserve traceability across heterogeneous agent stages, reproduce deterministic analyses, and alter its reasoning when exposed to previously accumulated structured scientific knowledge.

Resource or analysis	Input	Purpose	Runs executed	Verifiable output
Data resources, acquired and hashed by the deterministic host in the `data` phase				
Michaelis Table 1, epitope responses	extraction from the paper, 1.1 kB	peptide level T cell response counts		manifest entry `michaelis-table1-epitopes`
Michaelis Table 2, HLA ranks	extraction from the paper, 0.4 kB	published binding ranks, used as a control		manifest entry `michaelis-table2-hla`
Michaelis source data	published source data spreadsheet	per donor published values, read by the cytokine handler as a reproduction check		manifest entry `michaelis-source-data`
Michaelis Zenodo deposit	FlowJo workspaces, cytometry	donor level cytokine event counts		manifest entry `michaelis-zenodo-cytometry`
ALS GWAS summary statistics	GCST90027164, GRCh37, 197 MB	genetic association over the genome		manifest entry `als-gwas-gcst90027164`
IEDB NetMHCIIpan cache	predictions acquired once, cached for reproducible offline use	predicted HLA-II binding percentile ranks		manifest entry `iedb-netmhciipan-cache`
Deterministic analyses, run in Python in the `compute` phase				
A1 immuno-dominance	Michaelis Table 1	whether one epitope is dominant over the others, by Fisher tests with Holm correction and a bootstrap over rank one		`analysis/a1_immunodominance/summary.json`
A2 epitope clustering	peptide start positions, Michaelis Table 1	whether responding peptides cluster along the sequence, against an exact run distribution		`analysis/a2_epitope_clustering/summary.json`
A3 GWAS loci	ALS summary statistics	inflation and genome wide significant variants, with lead variants in two declared windows		`analysis/a3_gwas_loci/summary.json`
A4 HLA binding	IEDB cache, Michaelis Tables 1 and 2	reproduce the paper's seven published ranks, then correlate response with best predicted rank		`analysis/a4_hla_binding/summary.json`
Registered experiment handlers, run in the `execute` phase when arbitration selects them				
`hla_panel_compact_v1`	IEDB cache, allele panel of the cache	preregistered coverage of the local allele panel by a compact peptide set	3 of 5 (014, 015, 016)	`experiment-analysis/<EXP>/summary.json verification.json`
`ics_cytokine_skew_v1`	Michaelis Zenodo FlowJo workspaces, source data	preregistered re-analysis of the CD4+ IFN-γ / (IL-4 + IL-10) skew	5 of 5	`experiment-analysis/<EXP>/summary.json verification.json`

Table 2. Data resources, deterministic analyses, and registered computational experiments used in production runs. The host and the registered handlers execute every analysis listed here, and each emits a versioned, machine-verifiable artifact. All six resources and all four analyses were available in all five runs, so the last column counts handler executions only. Each manifest entry is referenced as `dataset:<id>@<digest prefix>` and each analysis summary carries a `result_digest`; both are omitted from the cells.

6. Results

The five production runs, `pl-prod-014` to `pl-prod-018`, follow the design of Section 5. Runs 014 to 016 start from an empty knowledge base (KB); runs 017 and 018 start from one accumulated snapshot, which carries 411 inherited mechanism edges. We call them the fresh-KB and the accumulated-KB replicates. Both accumulated runs started from the same snapshot of 1196 claims (identical `kb-snapshot.sha256`) and ended with 1287 and 1284 claims, after writing 91 and 88 of their own. Deduplication by canonical hash accounts for part of that difference: each wrote 5 computation claims, against 25 in run 016, because the compute phase recognised all 16 of its own claims as records the inherited knowledge base already held. Table 3 summarizes the five runs. With $n = 3$ and $n = 2$, results are reported per run and no inferential test compares the conditions.

			Experiments					
Run	Knowledge base at start	Claims	selected	executed	Mechanism edges / verified	Outcome	Cost (USD)	Wall time (min)
pl-prod-014	empty	109	6	2	2 / 1	`proposal`	14.24	35.2
pl-prod-015	empty	107	7	2	4 / 1	`proposal`	14.39	31.1
pl-prod-016	empty	104	5	2	6 / 2	`insufficient_evidence`	14.18	34.4
pl-prod-017	accumulated snapshot	91	4	1	12 / 1	`insufficient_evidence`	13.68	36.4
pl-prod-018	accumulated snapshot	88	2	1	13 / 1	`insufficient_evidence`	13.10	34.6

Table 3. The five production runs. *Claims* counts the claim records the run itself wrote (field `run_id` in `kb/claims.jsonl`), not the size of the knowledge base it read. *Experiments* reports how many of the designed experiments arbitration selected for execution and how many were executed. *Mechanism edges* is the longest causal chain cited by the final proposal, with the number of its edges whose quotation was found in the stored source (verified claims, §3.3). *Outcome* is the label the PI agent assigned to its own proposal, not a statement that a therapy was validated. Cost and wall time cover the twelve pipeline phases; the post-run `novelty` stage is excluded throughout.

6.1 End-to-end completion and proposal outcomes

All five runs completed the twelve core phases and passed the structural final-proposal gate. Each output contained all eight proposal sections and at least seven failure modes. However, structural completeness did not imply that the system endorsed a therapeutic intervention.

Run 014 carried the label `proposal` for three hypotheses that converged on antigen-specific tolerization, all still unresolved. Run 015 carried the same label, yet its prose stated that the confirmed evidence was insufficient to call any candidate a validated ALS therapy, and its connected mechanism held a single verified edge. Run 016 returned `insufficient_evidence` because the cytokine re-analysis did not support a TH1-conversion step its mechanism required, while the safety experiment that could have decided it had no registered handler. In run 017, two mechanisms relied on a CD80/IL-17A injury step contradicted by confirmed evidence, and the third, through IFN-γ, stayed unresolved. In run 018, the cited chain began with a contradicted link, and one hypothesis was rejected because its priming step targeted the wrong T-cell population.

The fresh-KB replicates therefore did not produce a deterministic high-level verdict: runs 014 and 015 carried the label `proposal`, whereas 016 returned `insufficient_evidence`. More importantly, the prose of run 015 declined to validate a therapeutic candidate despite its `proposal` label. For this reason, the categorical `outcome` field should not be interpreted as a sufficient summary of scientific support. The proposal text, evidence graph, and experiment results must be inspected jointly.

This behaviour is desirable in one limited sense: the system can terminate with insufficient evidence instead

of being forced to invent a therapeutic conclusion. However, the disagreement between the outcome label and the actual prose in run 015 exposes a schema-level weakness in the final summarization interface.

6.2 Replicate-level system behaviour

Table 4 summarizes the main run-level measurements. The three fresh runs generated three hypotheses each and proposed nine experiments each. The accumulated-KB runs also generated three hypotheses each and designed eight and nine, but fewer experiments were purchased and fewer executable analyses were triggered.

Metric	014	015	016	017	018
Outcome	proposal	proposal	insufficient	insufficient	insufficient
Sections citing contradicted or contested claims	0	0	0	2	1
Sections with citations judged unfaithful	2	1	1	1	0
Hypotheses generated	3	3	3	3	3
Experiments proposed / bought	9 / 6	9 / 7	9 / 5	8 / 4	9 / 2
Eligible bought / distinct executed / reused	4 / 2 / 2	6 / 2 / 4	4 / 2 / 2	1 / 1 / 0	1 / 1 / 0
Computed claims cited	9	11	7	5	2
Verification verdicts	51	38	44	28	30
Mechanism chain, edges / verified	2 / 1	4 / 1	6 / 2	12 / 1	13 / 1
`check_quote` exact rate, literature	0.94	0.86	0.78	0.84	0.94
`check_quote` exact rate, review	0.41	0.28	0.27	0.37	0.20
Claims cited by final proposal	26	27	27	24	36
Tool calls	343	348	344	308	346
Hypothesis diversity score	1.00	1.00	1.00	0.93	0.97

Table 4. Run-level measurements of the five production runs. A citation is *judged unfaithful* when this run's verifier marked the cited claim `faithful: false`, that is, the claim says more than its quotation; the final-proposal gate lists such citations as warnings. The `check_quote` exact rate is the share of judged quotation checks whose quotation occurs verbatim in the stored source, by the phase that requested them. Experiments *bought* are those the arbitrate phase selected for execution; *eligible* ones have a registered handler.

The diversity score is defined as one minus mean pairwise similarity of causal edges. It therefore measures structural edge overlap, not conceptual diversity of therapeutic ideas. This distinction mattered in practice: the fresh runs scored 1.00 while their proposal summaries nevertheless converged on closely related antigen-specific tolerization strategies. Consequently, the diversity metric should be read as graph-structural diversity, not semantic or therapeutic diversity.

The verifier returned substantial numbers of `insufficient` decisions across all runs. For runs 014 to 018, the confirmed/rejected/insufficient counts were respectively 19/13/19, 3/16/19, 12/4/28, 9/1/18, and 10/6/14. These distributions show that the verification layer does not simply confirm agent-generated relations. They do not, however, establish that the verifier is scientifically calibrated, because the verdicts were not independently audited by a domain expert.

Notably, the accumulated-KB runs did not yield more `proposal` outcomes: both returned `insufficient_evidence`, while two of the three fresh runs carried `proposal`. If a larger body of previously written claims is taken to support more confident conclusions, these two runs do not show it, and with two observations of a single snapshot the behaviour calls for further investigation.

6.3 Deterministic computational analyses

Four deterministic analyses were executed in the `compute` phase on the Michaelis et al. source data. Their numerical outputs were identical in all five runs, providing a deterministic reference point against which the variability of the LLM-driven phases can be contrasted. The analyses target peptide immunodominance, epitope clustering, genetic association, and HLA binding, all directly relevant to the C9orf72-centred starting point of the project [15].

All of these computations, and the two executable experiments below, re-analyse published observational data or predict binding in silico. They can reproduce, weaken or contradict an association reported in the source, but none can show that intervening on a mechanism changes the course of the disease. No hypothesis in any run received independent biological validation, and the results below are associational or predictive evidence, never causal evidence for a therapy.

Immunodominance (A1): the most frequent epitope stayed ranked first in 0.3456 of the bootstrap replicates, and none of its 20 pairwise comparisons survived correction, a recorded negative result, read as no single dominant peptide established. *Epitope clustering* (A2): 10 observed runs, that is contiguous blocks of responding peptides, $p = 1.4 \times 10^{-4}$, read as support for peptide-response clustering. *GWAS loci* (A3): $\lambda_{GC} = 1.117$, 483 genome-wide significant variants, lead $p = 9.98 \times 10^{-43}$ at 9p21 and 5.5×10^{-12} at the MHC, read as locus-level association only. *HLA binding* (A4): all 7 published ranks reproduced, 9 peptides with a predicted binder, Spearman $\rho = -0.194$ between predicted binding and response, a recorded negative result, read as predicted HLA binding not explaining immunodominance.

The HLA result is especially relevant to the intended therapeutic reasoning. Broad or predicted HLA-II binding is biologically relevant to the C9orf72 autoimmune observation [15], but the reproduced analysis did not support a simple relationship between predicted binding rank and observed immunodominance. The system therefore had computational evidence against using predicted HLA affinity alone as a sufficient explanation of peptide response strength.

Two registered analysis handlers could additionally be executed when selected by the arbitration stage: `hla_panel_compact_v1` and `ics_cytokine_skew_v1`. The cytokine handler returned the same numerical result whenever executed: the $CD4^+$ IFN-γ / (IL-4 + IL-10) ratio was lower in ALS than in controls (exact Mann-Whitney $p = 0.0113$, $n = 15$ vs 15). This is a case-control difference in a re-analysed cohort, not evidence that shifting the ratio would protect motor neurons. Its reproduction is also partial, and the failure has a structure. A donor counts as reproduced when its four recomputed responses equal a published row of the source data at two decimals. In every run, 7 of the 8 donors of the first of the four deposited ICS screens of the source study [15] (Zenodo, DOI 10.5281/zenodo.16749334) reproduced, and none of the 22 donors of the other three screens did. A disagreement confined to whole screens points to how the values of those workspaces are read, whether through gates, denominators or preprocessing, and not to the test, which is a fixed function of the per-donor values; which of these it is has not been established. The statistical procedure is fixed, but its result depends on the values it receives: applied to the 15 + 15 published rows instead of the recomputed ones, the same ratio, floor and exact test give $U = 49$, $p = 0.0075$ (medians 0.35 and 2.86, against 0.35 and 3.76 on the recomputed values; `scripts/ops/ics-published-skew.py`), so the direction of the difference does not hinge on the unreproduced screens. Until the cause of the disagreement is established, the $p = 0.0113$ result is read as agreeing in direction with the published claim on a partly reproduced dataset, not as an independent replication. The HLA panel satisfied its programmed panel condition in runs 014 to 016.

The deterministic results were reproducible, whereas their agentic interpretations were not perfectly stable. In runs 014 and 018, the interpretation of the cytokine-skew analysis omitted the limitation that only 7 of 30 donors reproduced the published values. The host-side evaluation detected these interpretations as unfaithful to the computational artifact. Runs 015, 016, and 017 preserved the limitation. This is a concrete example in

which a structured external check caught information loss introduced during LLM interpretation.

Reassessment remained conservative. Run 014 classified all four executed experiments as inconclusive. Run 015 retained the three cytokine-skew results and marked the three HLA results inconclusive. Run 016 produced one `weaken`, two `keep`, and one `inconclusive` reassessment; run 017 returned one `keep`; run 018 returned one `inconclusive`. No run established a final hypothesis verdict directly from an executed experiment. Decisive wet-lab experiments could be proposed but not executed by the current software, and some computational proposals lacked a registered handler.

6.4 Evidence grounding and verification behaviour

The two agent phases that most directly acquire textual evidence behaved differently (Figure 3). The exact-quotation rate of the `literature` phase ranged from 0.78 to 0.94, whereas the `review` phase ranged from only 0.20 to 0.41. In the latter phase most failed cases were absent quotations.

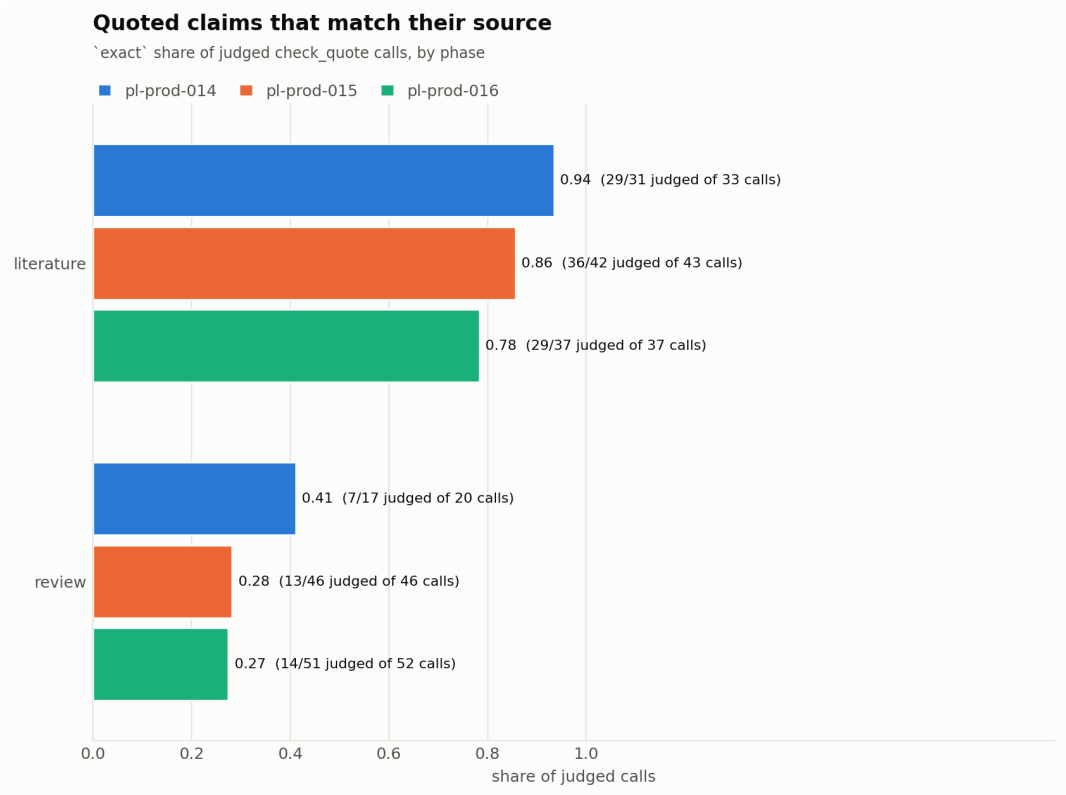


Figure 3. Share of judged `check_quote` calls whose quotation matched the stored source exactly, by phase, in the three fresh-KB replicates (`pl-prod-014` to `pl-prod-016`). In parentheses: exact matches over judged calls, and the total number of calls. The accumulated-KB runs, included in the ranges given in the text, are not drawn.

The difference concerns syntactic provenance only (§3.3); a separate verifier evaluates whether a proposed relation is supported by the quoted claim. This decomposition follows the broader motivation for evidence-grounded scientific agents: retrieval and citation are necessary but not sufficient for factual scientific synthesis [20]. It also avoids treating semantic proximity as equivalent to causal or evidential relevance, a limitation discussed in causal and graph-based RAG work [21, 22].

A second diagnostic concerned the scope of verification. In the shipped configuration, `only_on_run_`

`kb=true`, so the `verify` phase evaluates relations written in the current run. This was confirmed empirically: all 51, 38, and 44 verdicts in fresh runs 014 to 016 concerned same-run relations; the 28 and 30 verdicts in accumulated runs 017 to 018 also concerned same-run relations, despite the latter KBs containing thousands of inherited relations. Thus, accumulated knowledge can influence retrieval and hypothesis construction without being re-verified by the later run.

This asymmetry is central to interpreting the accumulated-KB experiments. The condition does not represent a system that repeatedly validates its entire growing scientific memory. It represents a system that reuses inherited claims while verifying only newly asserted relations.

In the fresh runs, hypothesis authors built on the evidence their own run had just collected: between 0.73 and 0.87 of the claims written by the `literature` phase were cited by hypothesis authors, and between 0.89 and 1.00 of the claims those authors cited came from that phase (Figure 4).

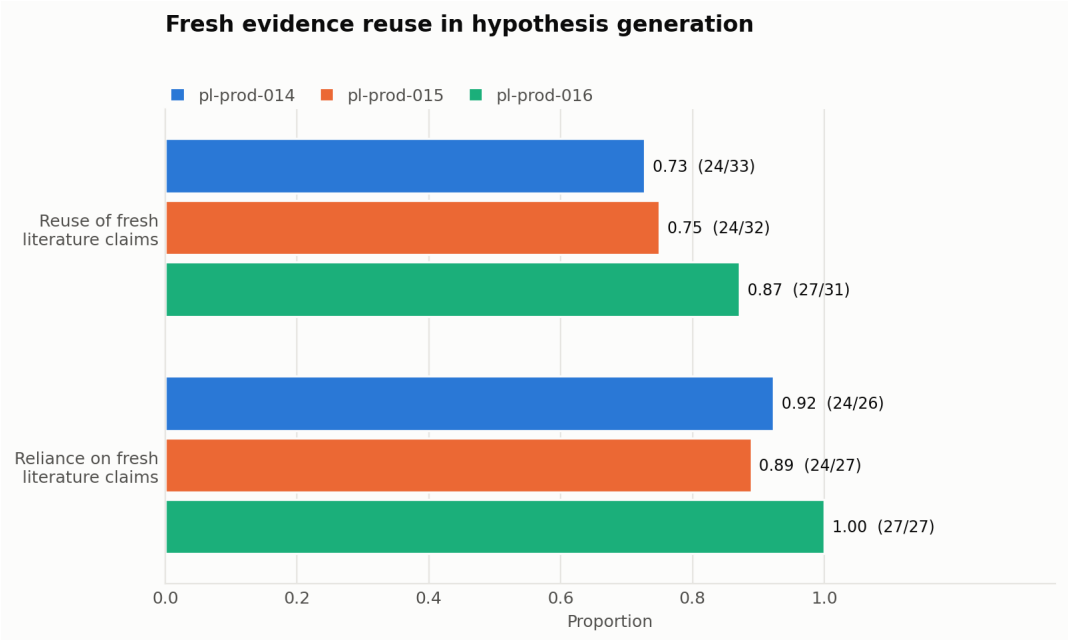


Figure 4. Fresh evidence reuse in hypothesis generation across the three fresh-KB replicates (`pl-prod-014`, `pl-prod-015`, `pl-prod-016`). Let F be the set of claims produced by the current run's literature phase and A the set of claims cited by hypothesis authors. Reuse $= |F \cap A|/|F|$ is the proportion of fresh claims linked by hypothesis authors; Reliance $= |F \cap A|/|A|$ is the proportion of author-cited claims that are fresh. Raw counts are reported in parentheses.

Four claims cited a web source that the final-proposal gate could not resolve to a stored paper. Checked by hand against the live pages on 2026-10-15, the pages exist and carry the quoted passages verbatim, but the claims add details that are not on the pages; the check is reported in the repository (`tab-web-sources-checked`).

6.5 Mechanism-chain growth does not imply evidential growth

The clearest diagnostic result concerns the causal chains used in the final disease-mechanism section. For each run, the reported chain is the longest simple path among the causal claims cited by the mechanism section (Table 5).

Of the agent-inferred edges, 0, 1, 1, 1 and 2 respectively (runs 014 to 018) were supported by a verified relation; the others remained unresolved or contradicted.

Run	Mechanism edges	Verified edges	Agent inferred edges	Inherited edges	Verified edge provenance
pl-prod-014	2	1	1	0	C-40, pl-prod-014
pl-prod-015	4	1	3	0	C-34, pl-prod-015
pl-prod-016	6	2	4	0	C-32, C-37, pl-prod-016
pl-prod-017	12	1	11	10	C-1035, pl-smoke-011
pl-prod-018	13	1	12	12	C-409, pl-prod-001

Table 5. Composition and provenance of the mechanism paths selected across production runs. Verified edges are causal claims whose provenance was verified against their stored source. Agent-inferred edges are causal claims introduced by agents without verified source provenance. Inherited edges were created in earlier runs and were already present in the initial knowledge base. The last column reports the identifiers and originating runs of the verified edges contained in each selected path.

The chain became much longer in the accumulated-KB runs, but the number of verified edges did not increase. Run 017 cited a 12-edge chain with one verified edge; run 018 cited a 13-edge chain with one verified edge. By comparison, the three fresh runs contained 2 to 6 edges with 1 to 2 verified edges.

The provenance columns of Table 5 make the effect more specific. In runs 014 to 016, all backbone edges were produced by the current run. In run 017, 10 of 12 backbone edges came from earlier runs; in run 018, 12 of 13 did. The single verified edge, the anchor that admitted the chain under the gate (§3.3), was also inherited in both accumulated runs. Thus, the increase in chain length was primarily a consequence of access to the accumulated KB, not of new verification performed by the current run.

This is the principal diagnostic finding of the experiment series: in both accumulated runs, the inherited claim graph let the PI agent cite a longer causal chain without any increase in verified causal support. The scope of this finding is narrow. The two accumulated runs started from the same snapshot, so they are two observations of one initial state, not of accumulation in general, and the groups of three and two runs support no estimate of an effect size. What the series shows is the mechanism by which a chain can lengthen without gaining support, not how often accumulation produces it.

The mechanism gate under which these runs were executed explains why this is possible. It checked that cited edges form a connected chain and that at least one edge is verified, but it required no minimum verified fraction and scored the plausibility of no inferred edge. Therefore, a 12-edge mechanism with one verified edge satisfied the same structural gate as a two-edge mechanism with one verified edge. This finding has since changed the gate: it now admits at most one unchecked edge, adjacent to a connected backbone whose every other edge is either quote-verified or supported by a verified relation. The five runs reported here predate that change and none has been repeated under it, so the tightened rule is reported as a consequence of this series, not as a result of it.

6.6 Novelty assessment

The post-pipeline novelty stage attempted to determine whether each generated hypothesis was already anticipated in the literature. No successfully assessed hypothesis received a supported `no_prior_art_found` verdict.

For runs 014 and 015, one hypothesis was classified as anticipated and two as partly anticipated. Runs 016 and 017 classified all three as partly anticipated. The stage read full texts only in runs 014 and 015, three and four works respectively. The 018 novelty evaluation failed because the external search dependency returned HTTP 429 responses for essentially all search requests. Consequently, the absence of a novelty verdict in run

018 is an infrastructure failure, not evidence of novelty. The `partly anticipated` verdicts of runs 016 and 017, reached without reading any full text, therefore measure the behaviour of the automated assessor more than the novelty of the hypotheses, and none of these verdicts shows that a hypothesis lacks originality.

These findings are consistent with the difficulty of evaluating machine-generated hypotheses in open-ended scientific domains, where novelty, plausibility, relevance, and testability are separate dimensions and no complete ground truth exists [1]. They also reflect a classical limitation of literature-based discovery: producing a previously unconnected path is not by itself sufficient to establish that a scientifically meaningful hypothesis is genuinely novel [9].

6.7 Computational cost and latency

The twelve core phases cost USD 14.24, 14.39, 14.18, 13.68 and 13.10 for runs 014 to 018, with wall times of 35.2, 31.1, 34.4, 36.4 and 34.6 minutes and 35, 35, 34, 28 and 27 agent launches; the post-run novelty stage is excluded. Across the five runs, the twelve-phase pipeline cost USD 69.58. The separate novelty stage added at least USD 6.77 for the final attempts, with additional failed/retried calls in runs 014 and 018. The accumulated-KB runs launched fewer agents, mainly because fewer hypotheses progressed to verification, execution, and reassessment. Given the small sample, the lower cost of runs 017 to 018 should be treated descriptively, not as evidence that KB accumulation reduces computational cost. These amounts depend on the agent configuration a run used, not on the workflow alone: the number of agents, their models, their hop limits and their caching are declared phase by phase in `configs/pipeline.yaml`, and each USD figure applies a dated token price table (`configs/tokens.yaml`) to the tokens those agents consumed. A different team of agents, or a later price list, yields different figures; the token counts the archives record are the price-independent measurement.

6.8 A representative trace

Figure 5 follows one path through run `pl-prod-016`, from a verified literature claim to the final PI judgement, to show how verified evidence, agent inference and computational experiment remain distinguishable within a single case.

6.9 Summary of findings

The end-to-end system successfully executed the complete 12-phase workflow in all five production runs and generated structurally valid final proposals. The deterministic biological analyses were exactly reproducible across runs, whereas hypothesis generation, arbitration, interpretation, and final outcome were variable. The system also demonstrated several useful forms of epistemic restraint: it frequently returned insufficient verification, could decline to endorse a therapy, and in one run explicitly rejected a hypothesis.

The main weakness emerged where persistent memory met causal synthesis: in the two accumulated-KB runs the cited chain was longer, mostly inherited, and no better verified. Section 7 interprets these observations.

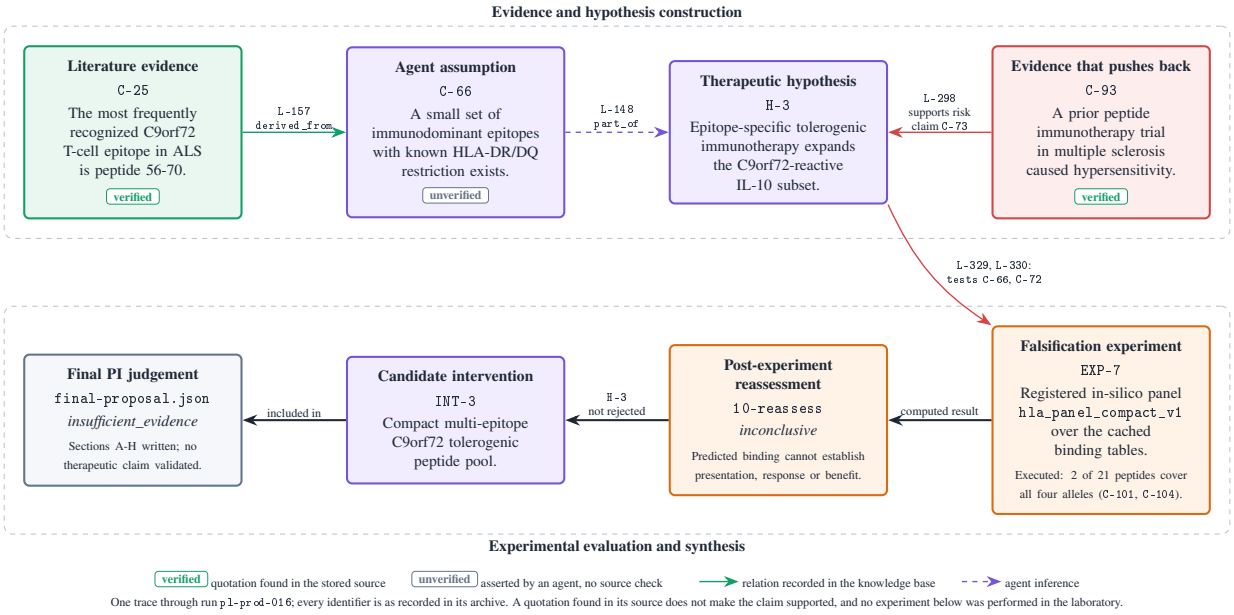


Figure 5. Representative scientific trace from run `pl-prod-016`: one path through the run, not its whole reasoning graph. The diagram follows a literature claim whose quotation was found in its source (`C-25`) through an agent-generated mechanistic assumption (`C-66`), a therapeutic hypothesis (`H-3`), counter evidence (`C-93`, which supports `C-73`, a risk assumption of that hypothesis), a registered in-silico falsification experiment (`EXP-7`), post-experiment reassessment, a candidate intervention (`INT-3`) and the final PI judgement. Solid arrows denote relations stored in the knowledge base, dashed arrows agent inference.

7. Discussion

The experiments suggest that the main challenge in autonomous scientific discovery is not hypothesis generation itself, but controlling how generated reasoning becomes accepted scientific state. The system consistently completed the full workflow, generated competing hypotheses, proposed experiments, executed available computational analyses and produced a structured final proposal. At the same time, repeated runs exposed a clear separation between workflow completeness and epistemic strength. A proposal can be structurally complete while the mechanism supporting it remains only weakly verified.

This distinction is important because current AI scientist systems often improve scientific reasoning through repeated generation, critique, ranking or refinement [6, 13, 19]. Such mechanisms can improve candidate quality, but they do not by themselves establish correctness. Intrinsic model based correction can even decrease reasoning accuracy when reliable external feedback is absent [7]. Our results support a related conclusion at the workflow level: additional reasoning and accumulated state can increase the amount and connectivity of scientific argument without proportionally increasing independently checked evidence.

7.1 Accumulated knowledge is reused but not re-verified

The clearest result concerns the persistent knowledge base: in both accumulated runs the cited mechanism chain grew longer, mostly from inherited edges, while its verified edges did not increase (§6, Table 5). Two runs sharing one initial snapshot cannot show how general this is; they show that the present verification policy permits it.

This is not merely a graph size effect. The verifier was configured to inspect relations created by the current run. Earlier relations could therefore be retrieved and reused by later agents without being reconsidered by the verifier. The architecture consequently gives retrieval a wider temporal scope than verification. In the accumulated condition, old inference can remain available indefinitely, whereas scrutiny remains local to the new run.

The result identifies a concrete failure mode of this design, and plausibly of persistent agentic knowledge systems more broadly, although two runs cannot establish the latter. An append only memory preserves provenance and prevents silent rewriting, which is desirable for reproducibility. However, persistence also allows weakly supported inference to acquire structural importance simply because later agents reuse it. Knowledge accumulation should therefore not be interpreted as evidence accumulation.

A stronger design would make epistemic maintenance an explicit operation. Retrieval of an inherited causal path should trigger verification of the relations required by the current argument, or at minimum expose the verified fraction and provenance of that path to downstream agents. Causal and graph based retrieval approaches similarly motivate moving beyond semantic relevance toward explicit relational structure [21, 22], but the present results indicate that structured retrieval alone is insufficient. The epistemic status of retrieved relations must also constrain their use.

7.2 External feedback matters more than agent agreement

Four design decisions can be read against what they stopped in the five runs and what passed them anyway, summed from the runs' scorecards. First, an agent's relation counts as evidence only after the verification agent confirms it and finds the claim no broader than its quotation: of 191 verdicts, 40 relations were rejected, 17 were confirmed but discarded as unfaithful, and 36 became evidence; once admitted, however, nothing is re-verified, so inherited relations entered later chains unchecked (§6). Second, a reviewer's unsupported objection is kept as a question: 189 objections were held back from the contradiction count, at the cost, not

audited, that a correct objection without a quotation weighs nothing. Third, an executed analysis is interpreted against a rule fixed before its result was known: the check flagged the cytokine readings that dropped the 7-of-30 donor caveat in run 014, for three interpretations, and in run 018, for one, but the flag does not block. Fourth, the final-proposal gate blocked every unknown record and every experiment claimed without a receipt, and only warned on the rest: unfaithful citations in four runs, contradicted or contested claims in two, a 13-edge chain with one verified edge, and run 015's `proposal` label against its own prose.

What held is what the checks block; what passed is what they only report. Agreement between agents was never what admitted a relation: the verifier and the quotation check were. That is the sense in which external feedback matters more than agent agreement in this system, and it is also why the errors that survived are errors of degree, a claim stretched beyond its quote, rather than of invention. Literature agents such as PaperQA2 show that agentic search and citation traversal can improve retrieval substantially [20]; these results suggest coupling such retrieval to provenance-preserving checks and making the diagnostic ones blocking where their false-positive rate allows.

This separation also explains why an `insufficient_evidence` outcome should be considered a valid product of the workflow rather than a failure. Three of the five runs returned it, and a fourth reached the same conclusion in prose (§6). An autonomous discovery system that is forced to terminate with a therapeutic candidate would reward coherence over evidential restraint. For this project, the ability to decline a candidate is therefore part of the scientific function of the system.

7.3 Computation provides the strongest reproducible feedback

The deterministic analyses behaved differently from the generative stages. The four predefined analyses returned the same numerical results across all five runs, whereas hypothesis selection and final interpretation varied. This gives the workflow a useful epistemic anchor: computational results can be reproduced independently of the stochastic language model trajectory.

However, the experiments also expose the present boundary of autonomy. Only two experiment handlers were executable, so the system could formulate decisive biological experiments without actually performing them. Reassessment could therefore keep, weaken or mark hypotheses as inconclusive, but it could not close the scientific loop for hypotheses requiring new wet laboratory evidence. This is consistent with the broader AI scientist literature, where strong claims ultimately rely on external experimental validation rather than internal model consensus [6].

The practical implication is that expanding the number of agents is less important than expanding the set of trustworthy feedback channels. A future version should prioritise executable analyses, pharmacological and biological tools, and mechanisms that convert external results into explicit updates of hypothesis status. The current implementation demonstrates the orchestration substrate for such a loop, but not yet autonomous therapeutic discovery.

7.4 Implications for autonomous scientific workflows

Taken together, the experiments support a narrower claim than complete autonomous discovery. The system demonstrates that literature search, hypothesis generation, adversarial review, evidence verification, experiment design, computational execution and final synthesis can be integrated into one workflow whose deterministic analyses are reproducible and whose intermediate scientific state remains inspectable. This satisfies the architectural objective of moving beyond isolated chatbot interactions toward an agentic scientific ecosystem.

The stronger scientific claim remains unresolved at the moment. No run established a therapeutic intervention with sufficient verified causal and experimental support. The main contribution of the experiments is therefore

diagnostic: they show where an apparently successful AI scientist can become epistemically weaker than its output suggests.

For an autonomous scientific system, the relevant scaling target is therefore not the number of hypotheses, agents or graph edges. It is the amount of independently checkable evidence that survives adversarial scrutiny and can support the next experimental decision.

8. Limitations

Despite its functional implementation, the proposed system has limitations concerning scientific validity, verification, autonomous experimentation, and reproducibility.

Epistemic reliability. The knowledge base distinguishes verified claims from agent inferences and computes epistemic status from verified evidence relations. However, verification stops at semantic entailment (§3.3): the workflow does not independently establish the empirical validity of any claim. The mechanism graphs represent causal assertions extracted or proposed by agents, rather than causal structures identified from experimental data. The mechanism gate in force for these runs admitted a chain with a single verified edge (§6); it was tightened afterwards, and no run reported here used the tightened rule. Chains also join only where consecutive claims reference the same entity record, and whether two mentions reach one record rests on shallow lexical normalization and named rewrite rules; the mechanism tool applies no deduplication groups. A chain can therefore break where one entity is named in two ways, or join where two entities share a name. Learned or ontology-based entity resolution is left to future work.

Verification and agent dependence. Independent reviewers and deterministic quotation checks mitigate some errors but do not guarantee faithful reasoning. Quotation matching was markedly less reliable in review than in literature collection (§6). Furthermore, independently generated hypotheses may share biases arising from common models and evidence. Critique alone cannot guarantee correction without reliable external feedback [7].

Experimental scope. Four deterministic analyses were reproduced across the five runs, but autonomous execution was limited to two registered computational procedures. Decisive biological experiments remained proposals, without experimental execution or validation. The `reassess` verdict remains a report. Closing this gap requires the host to write the relation from the handler's preregistered interpretation of the outcome. Consequently, therapeutic efficacy, patient stratification, and safety cannot be established from the generated proposals.

Reproducibility and evaluation. Three runs initialized with empty knowledge bases produced different final outcomes despite identical configurations and computational results. The small number of runs precludes statistically robust conclusions about comparative performance. Moreover, the novelty assessment was incomplete, citation faithfulness was not comprehensively audited across these runs, and evidence inherited from previous runs was not independently reverified.

9. Conclusion

This project investigated the question: *Can a multi-agent system autonomously transform a biomedical observation into auditable, mechanistically explicit, and falsifiable therapeutic hypotheses for amyotrophic lateral sclerosis (ALS)?*

The answer is a qualified yes for the question as posed, which concerns generating hypotheses, not demonstrating that a therapy works. A multi-agent platform integrated literature, deterministic computation, competing

mechanistic hypotheses, adversarial review and experimental design into one workflow. Every proposal it produced refers to knowledge-base records that a reader can inspect, which lets the reader distinguish source-grounded evidence, re-checked computations and unverified agent inferences; a traceable record, however, is not a grounded claim, and four claims cited web pages that the final gate could not resolve to a stored source (§6). Each proposal also specifies in advance the experiments that could falsify it, but whether those experiments would discriminate between the competing mechanisms was not assessed by domain experts. The qualifications concern the quality of those hypotheses: their mechanisms rest largely on agent inference, with one or two verified edges in each cited chain (§6), and the automated novelty assessment found none of them free of prior art. That assessment is incomplete: it read full texts only in runs 014 and 015, and it failed in run 018. It therefore describes the assessor's behaviour more than the originality of the hypotheses, which it neither establishes nor rules out. Therapeutic efficacy is a separate question that this work does not address: the computational analyses were executed, but no proposed therapeutic intervention underwent experimental efficacy testing, and no result bears on whether any of them could modify the course of ALS.

Across five runs the platform produced falsifiable proposals, and in three of them it declined to propose a therapy. The same runs exposed where its checks stop: a quotation found in its source can still be stretched beyond what it says, inherited relations are not re-verified, and the outcome label can disagree with the proposal's own prose.

A major challenge is therefore not merely improving the reasoning capabilities of individual agents, but providing the infrastructure necessary for scientifically reliable discovery. This requires access to comprehensive biomedical research databases, specialized computational biology tools, and experimental resources capable of evaluating candidate mechanisms. In particular, integration with laboratory workflows would enable agents to move beyond literature based hypotheses toward iterative cycles of experimental testing and evidence driven revision.

Human expertise remains essential to this process. Incorporating domain experts through a human in the loop architecture would provide independent scientific judgment, oversight of experimental decisions, and interpretation of results that cannot be reliably delegated to language models alone.

In conclusion, this work demonstrates the engineering feasibility of a platform that generates research proposals and exposes them to verification, and it identifies the scientific and engineering effort still required before such a platform could discover an effective therapy. The longer term opportunity lies in connecting agentic reasoning, computational and laboratory experimentation, and human expertise into a continuous scientific discovery process.

9.1 Code and data availability

The implementation, its configuration and the scripts that acquire the third-party inputs are available at https://github.com/gianfa/als-ai-scientist. The archives of the production runs reported here, including their knowledge bases, phase reports, scorecards and final proposals, are deposited at https://doi.org/10.5281/zenodo.23370543. The third-party inputs are not redistributed: their records and licences are listed in `LICENSES.md`, and they are acquired by the scripts documented in `docs/data-acquisition.md`.

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